

Trust Me, I'm a Certificate Authority

PyCon PT 2026 Tutorial

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DISCLAIMER

THIS IS TEACHING MATERIAL, NOT SECURITY SOFTWARE.

IT CUTS CORNERS ON PURPOSE.

SEVERAL EXERCISES ARE DELIBERATELY BROKEN.

REAL SECURITY IS A GREAT DEAL MORE THAN THIS.

USE NONE OF IT FOR ANYTHING THAT MATTERS.

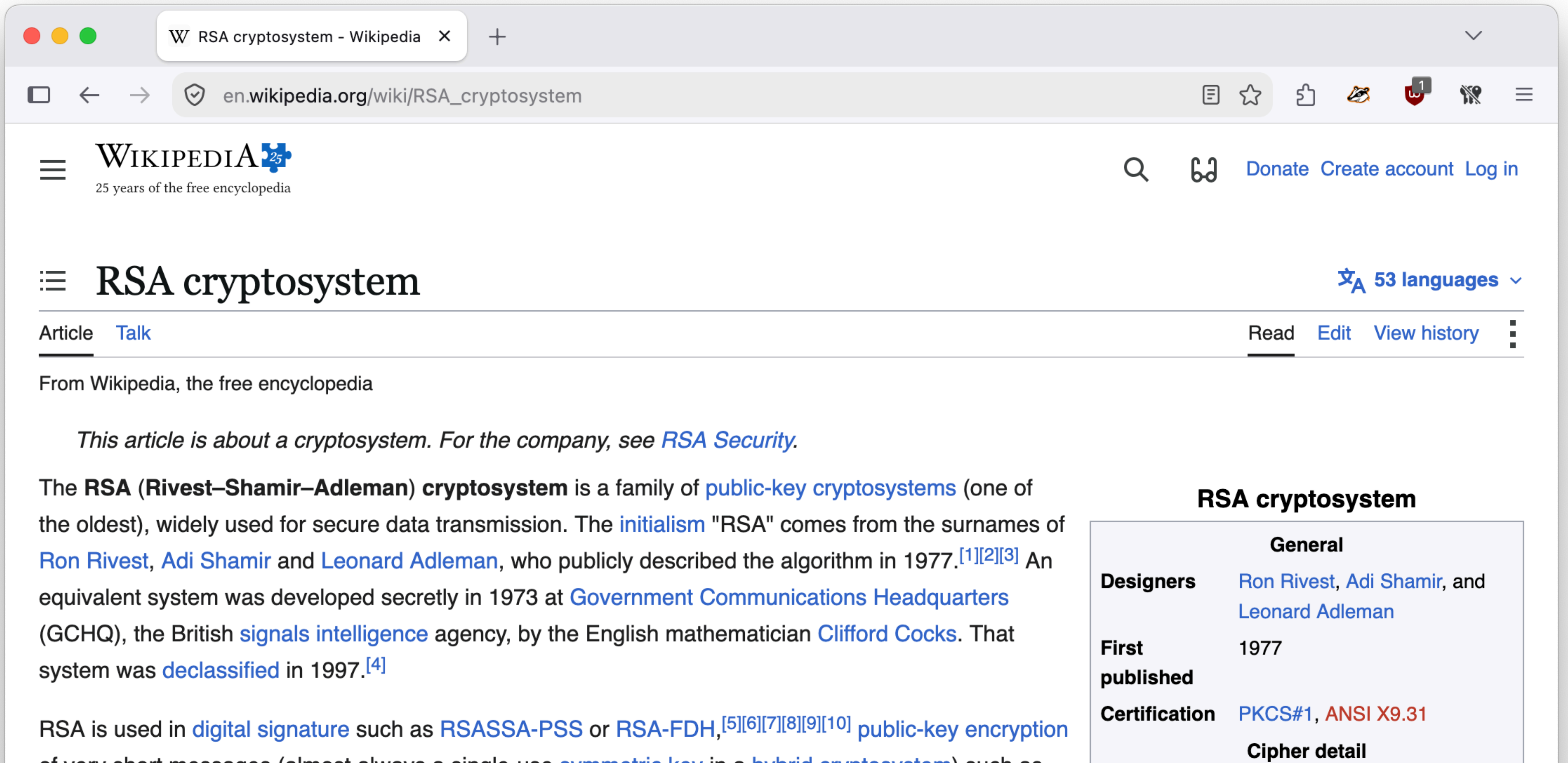
NEVER TRUST A CA YOU BUILT HERE.

Trust Me, I'm a Certificate Authority

- **RSA** Keys, encryption, and signatures.
- **Certificates** Binding keys to identities (yay X.509)
- **Cert Authorities** Becoming a Certificate Authority
- **TLS with HTTPS** Real HTTPS with a client that trusts.
- **MITM** Interception: where trust actually lives
- **Break Stuff** Certificates that fail, checks that catch them

RSA (Rivest–Shamir–Adleman)

RSA (Rivest–Shamir–Adleman)

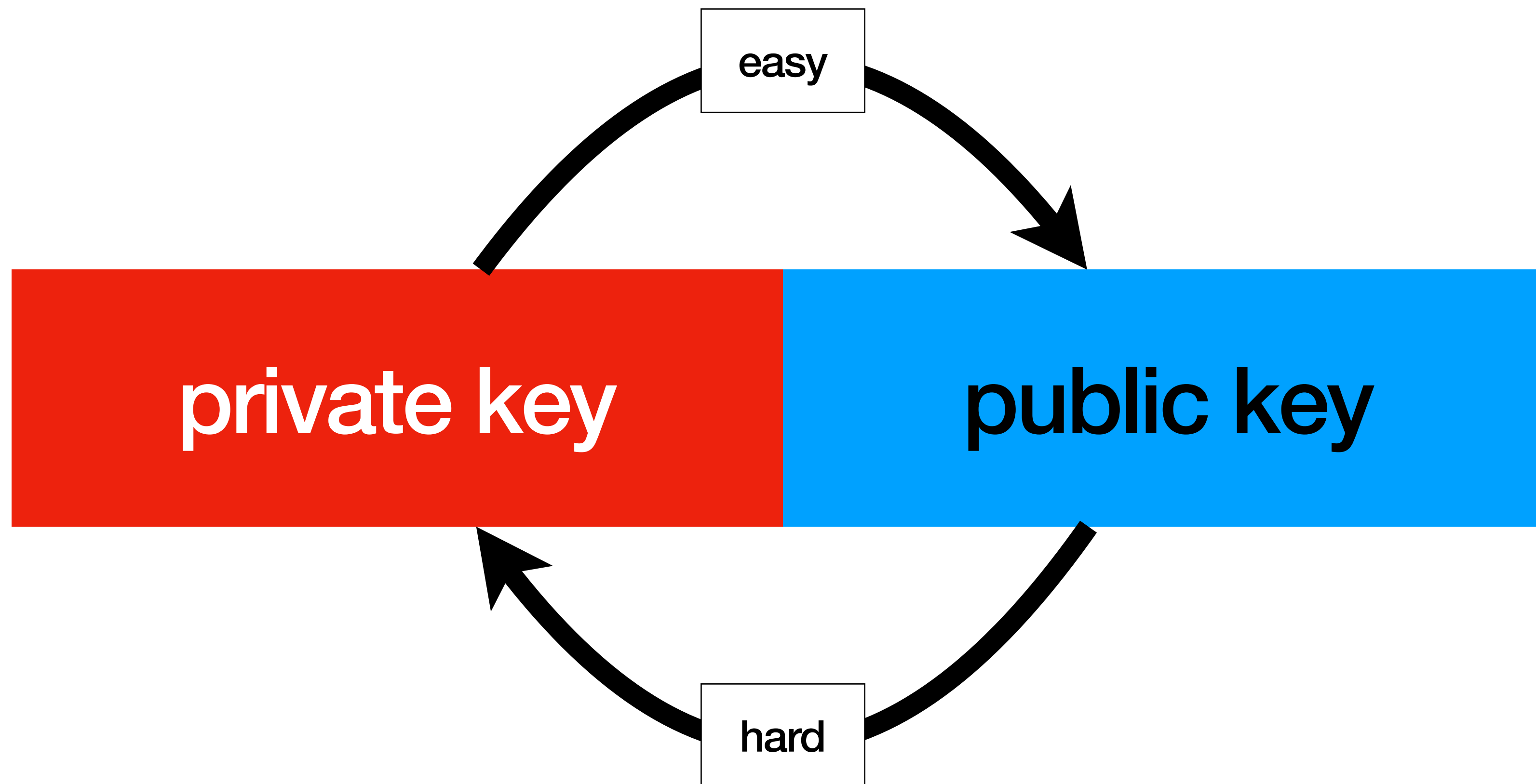


RSA key pair

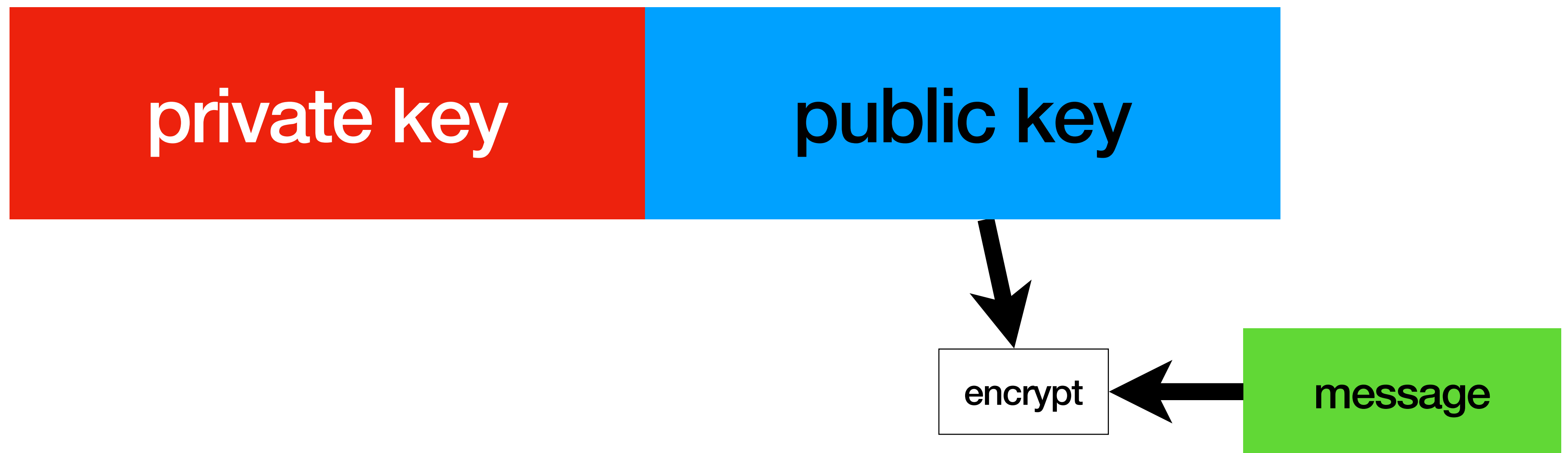
private key

public key

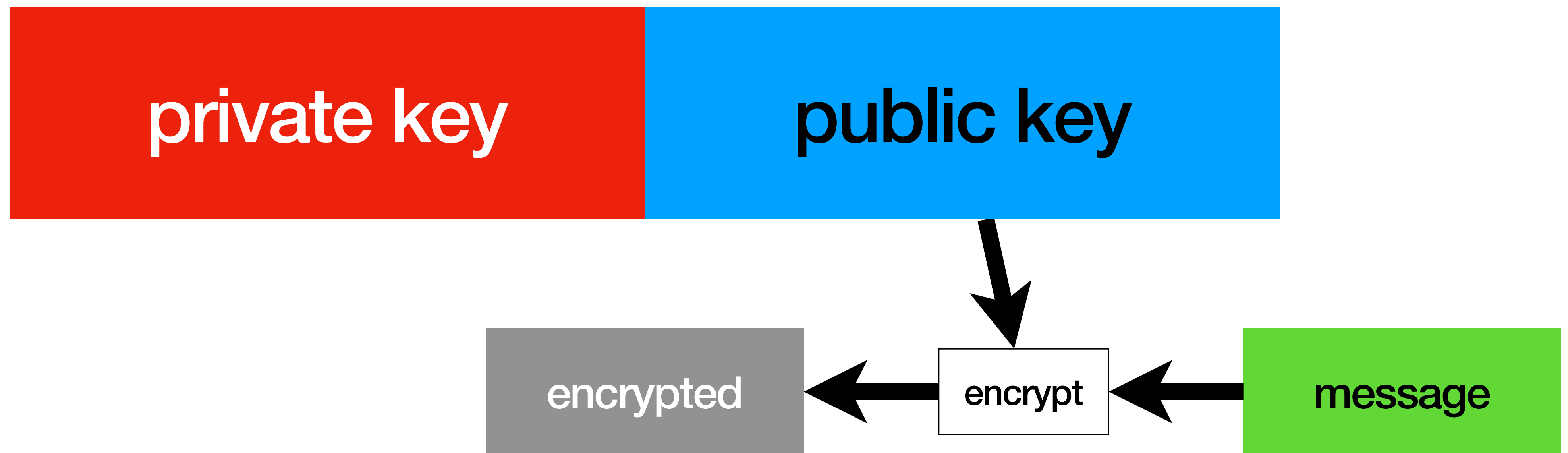
RSA key pair



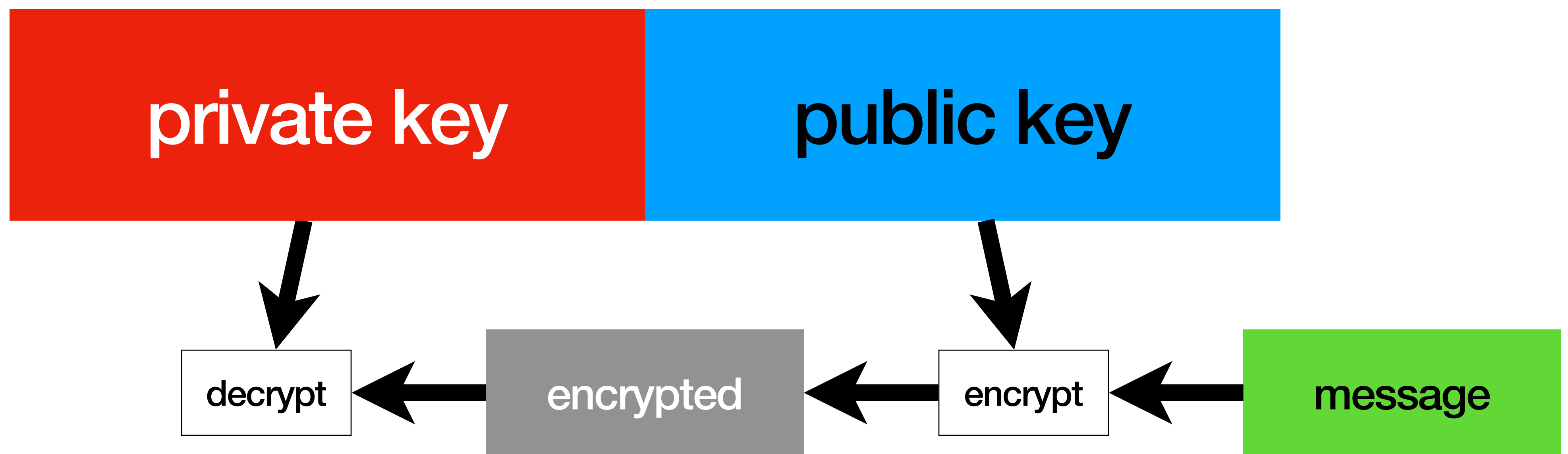
RSA operations



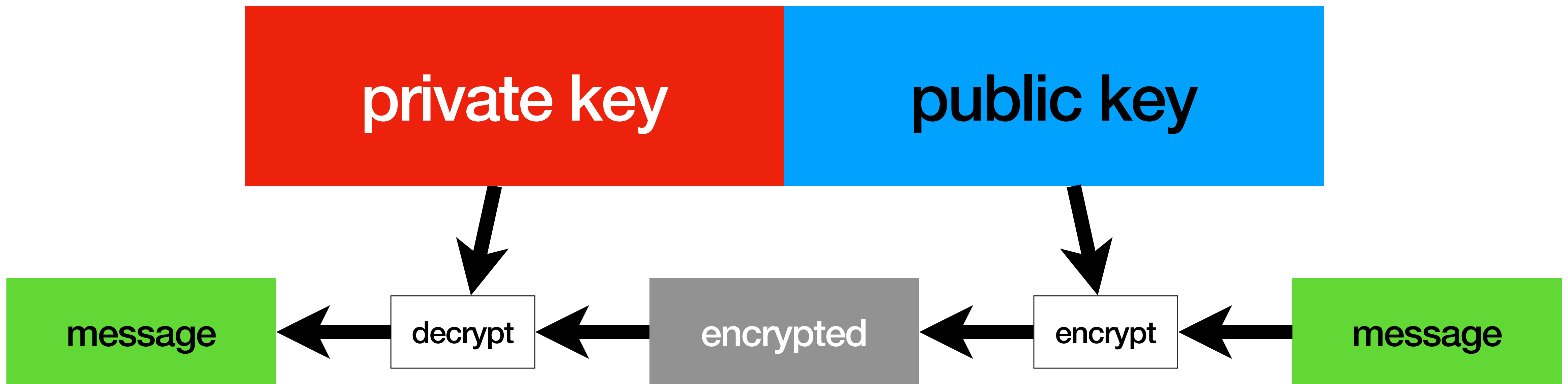
RSA operations



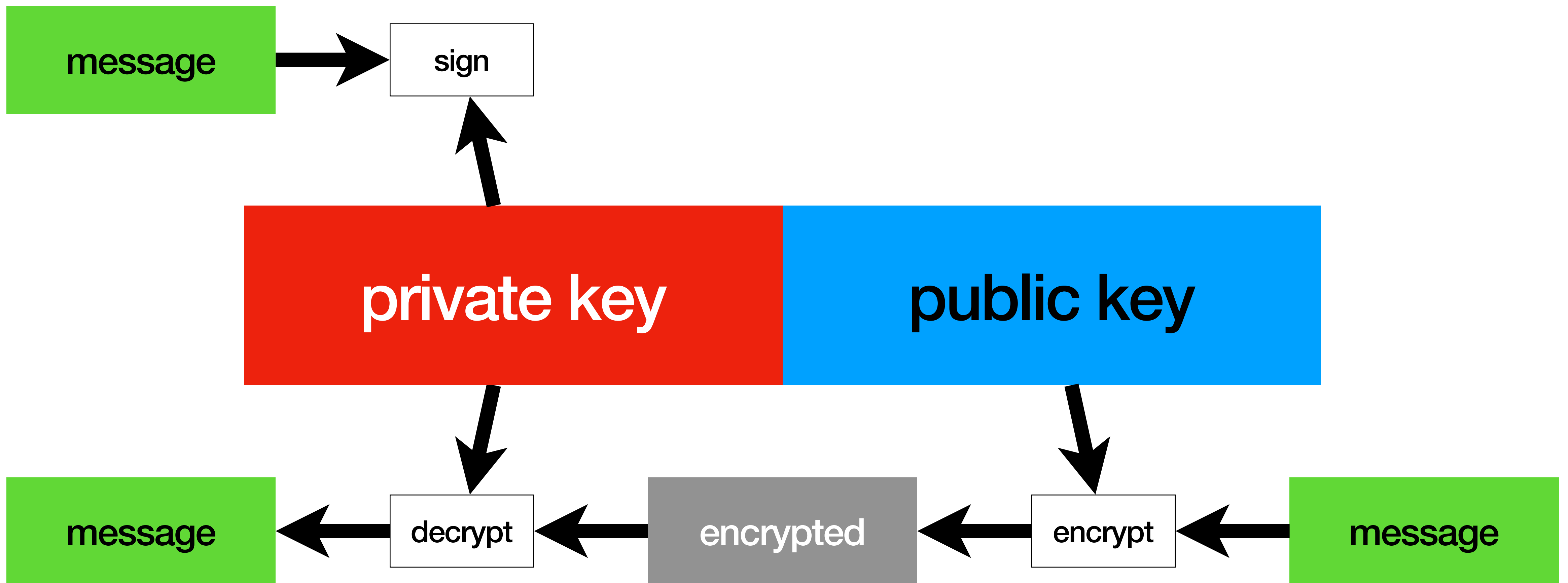
RSA operations



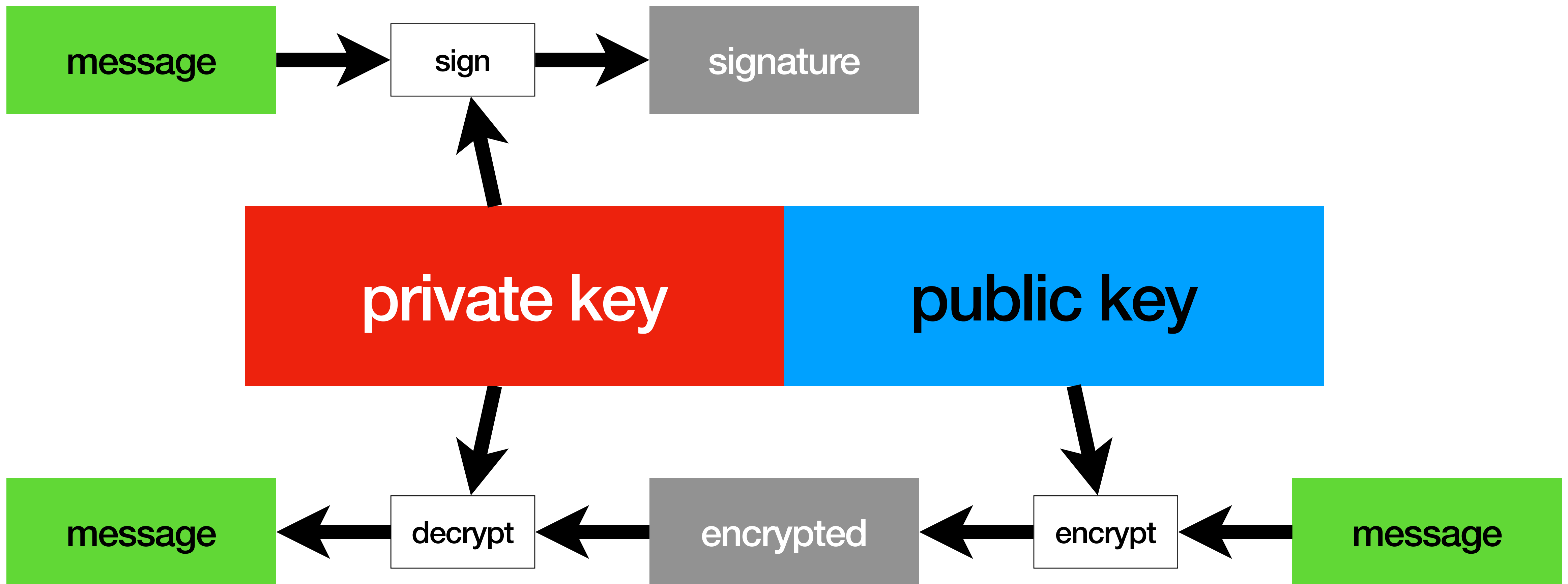
RSA operations



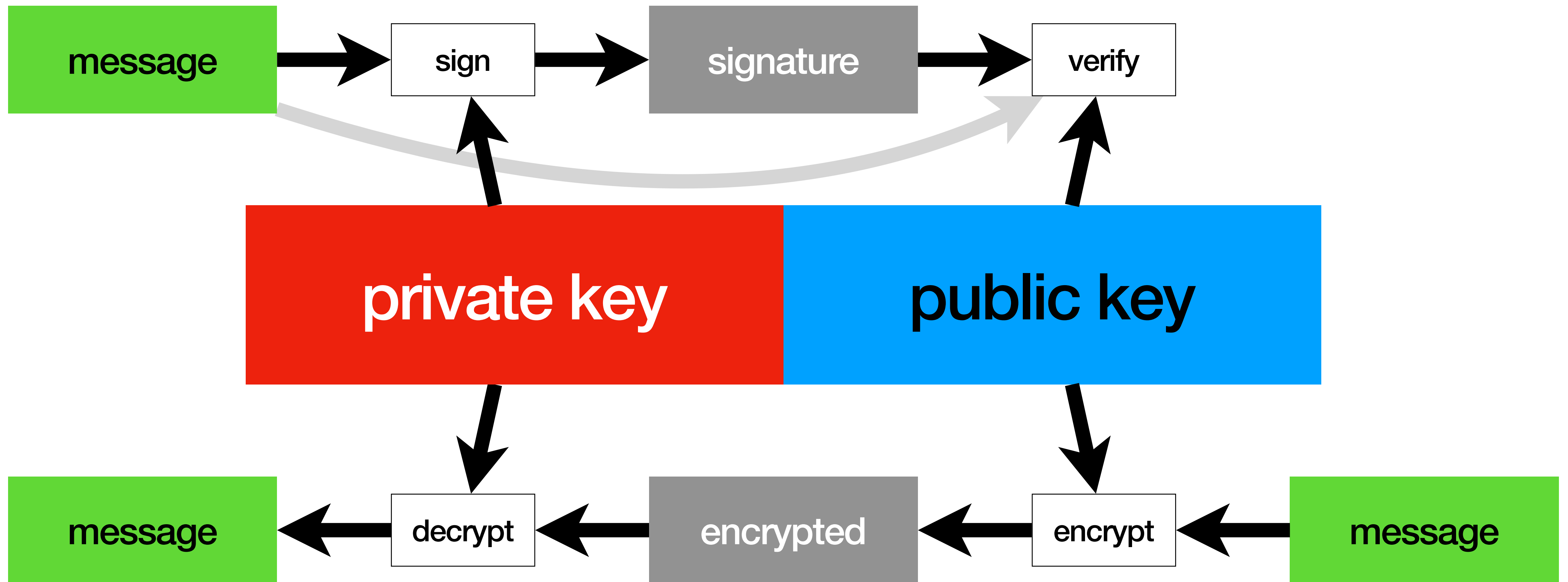
RSA operations



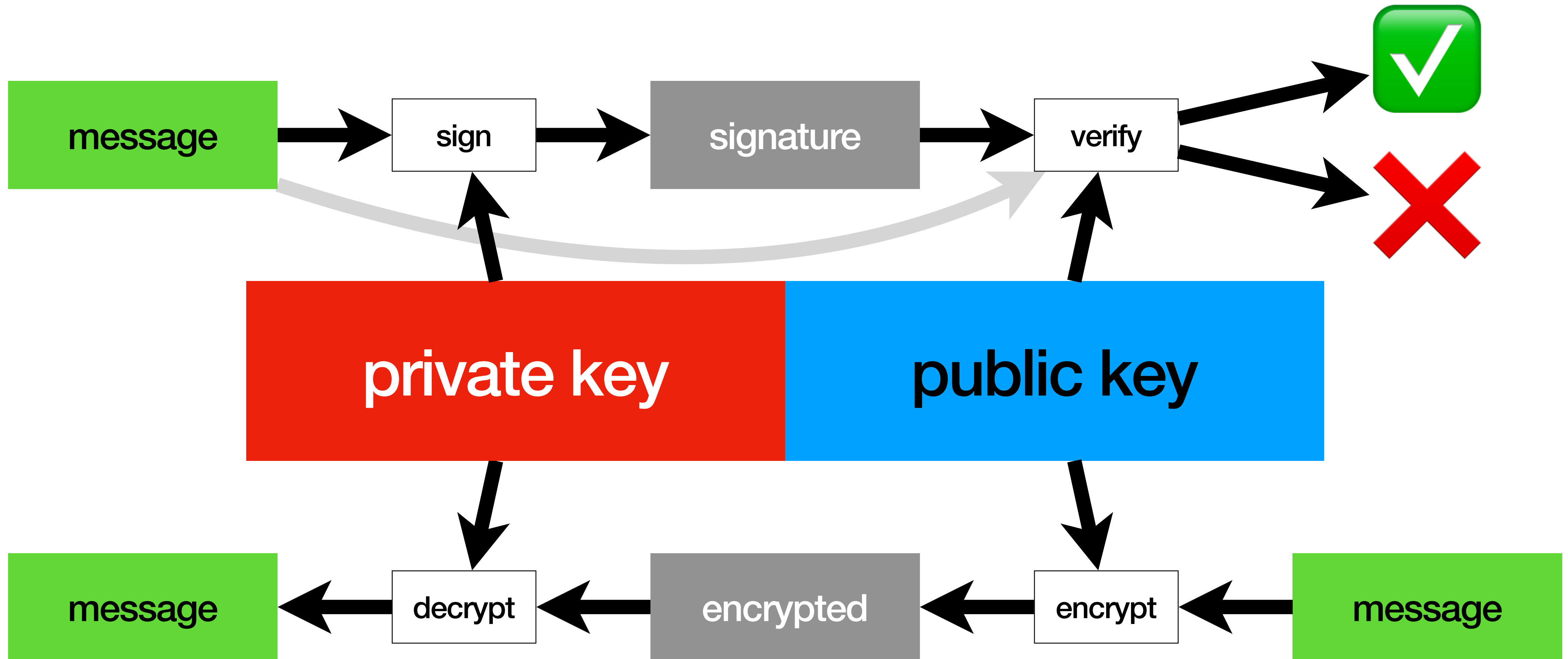
RSA operations



RSA operations



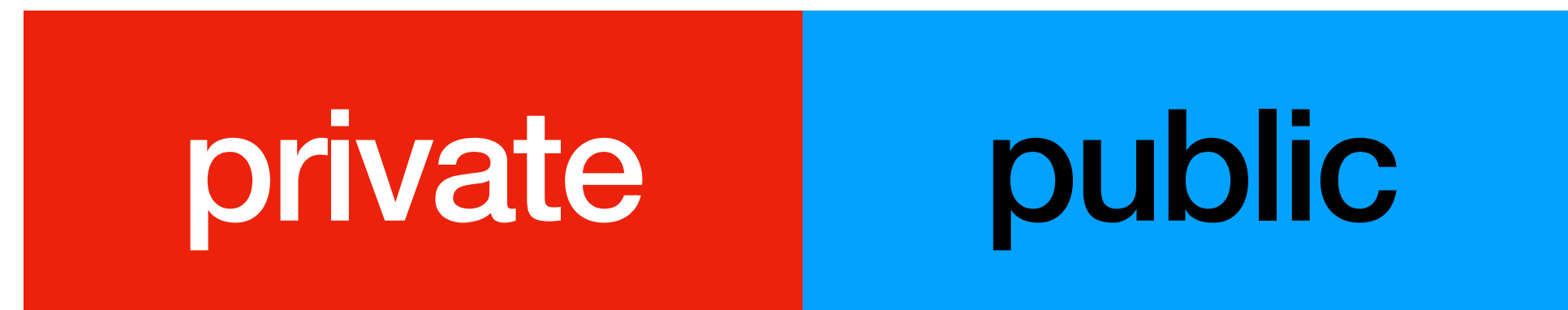
RSA operations



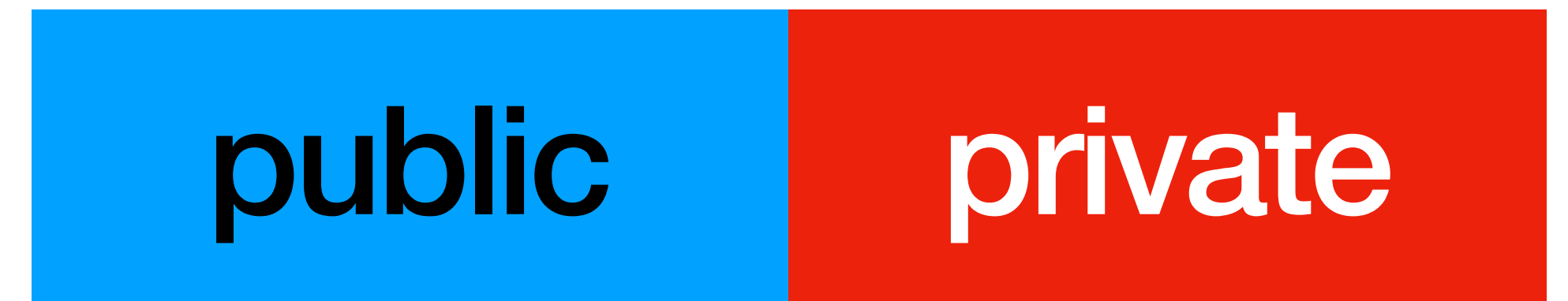
RSA private communication

RSA private communication

Alice

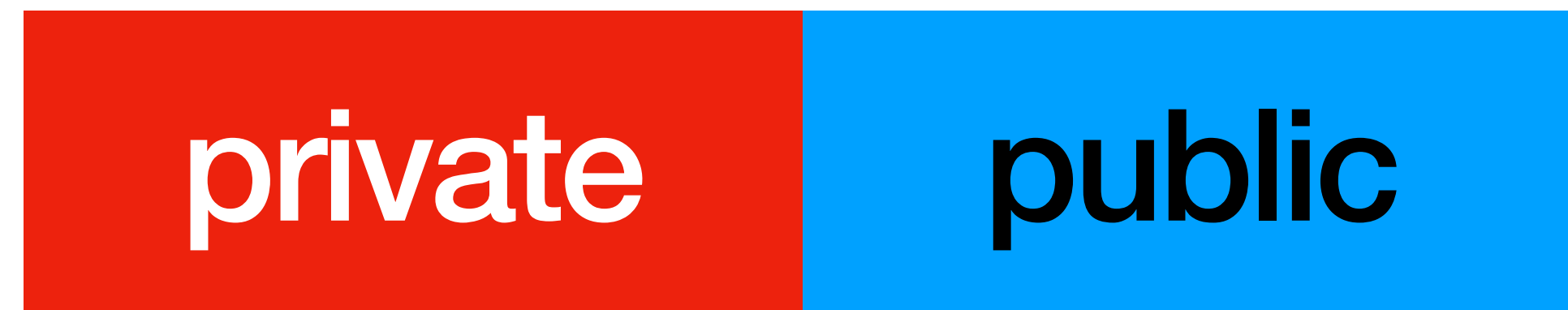


Bob

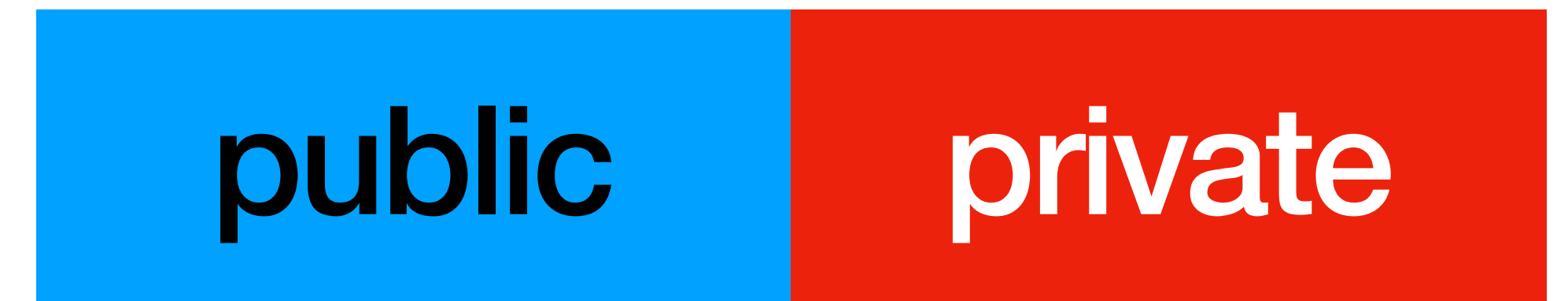


RSA private communication

Alice

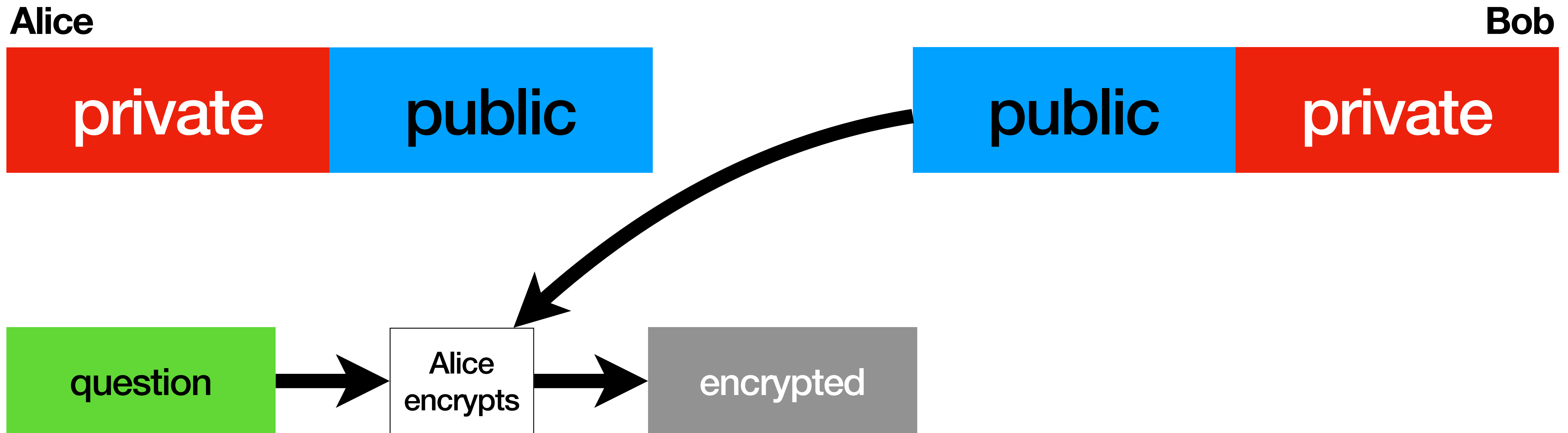


Bob



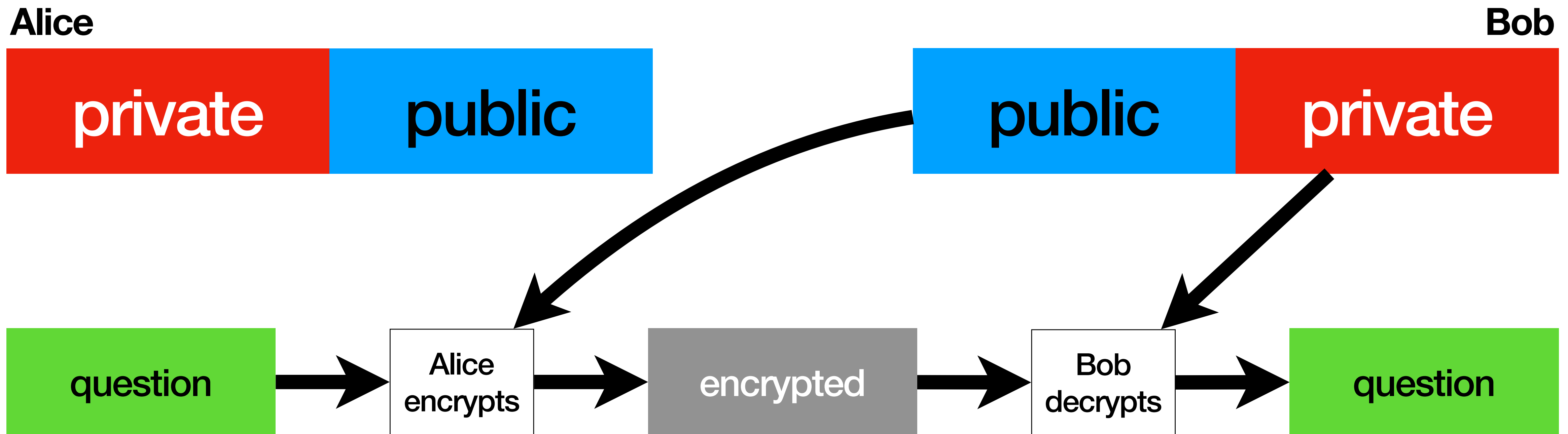
Alice sending a private question to **Bob**

RSA private communication



Alice sending a private question to **Bob**

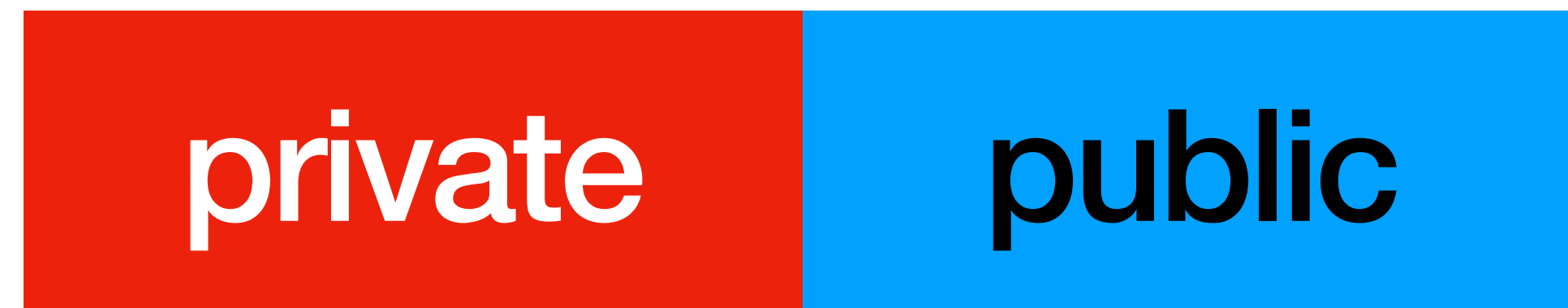
RSA private communication



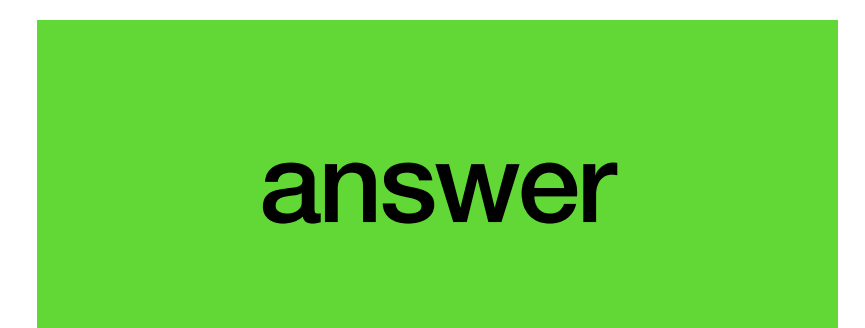
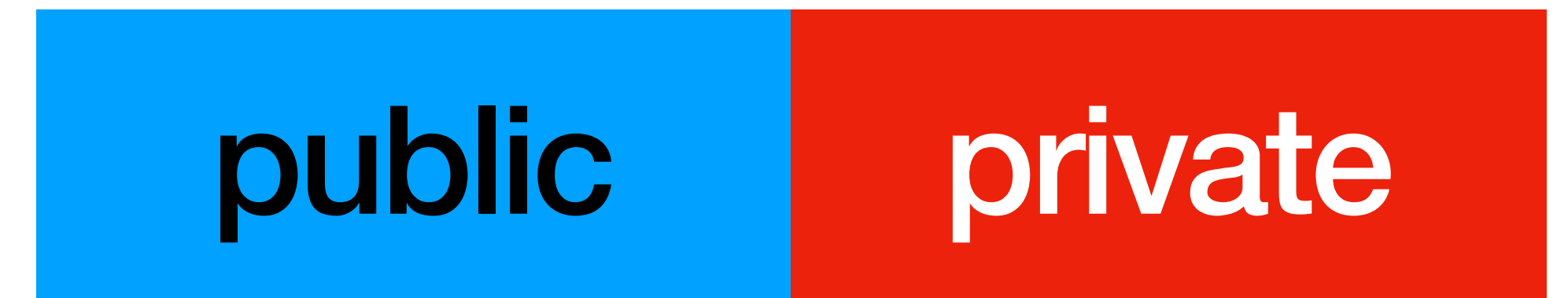
Alice sending a private question to **Bob**

RSA private communication

Alice

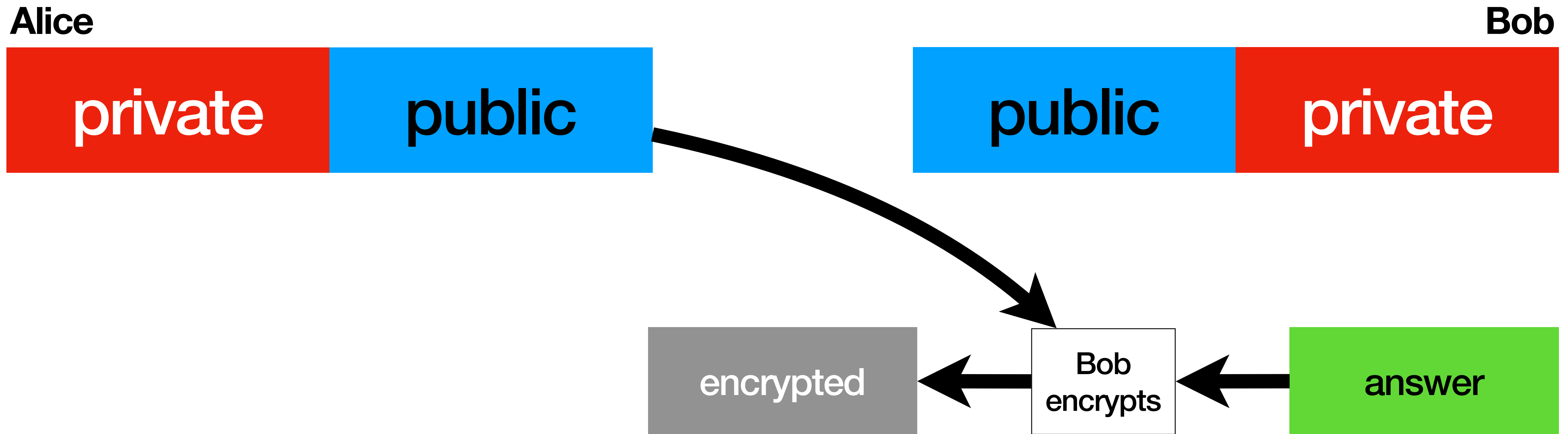


Bob



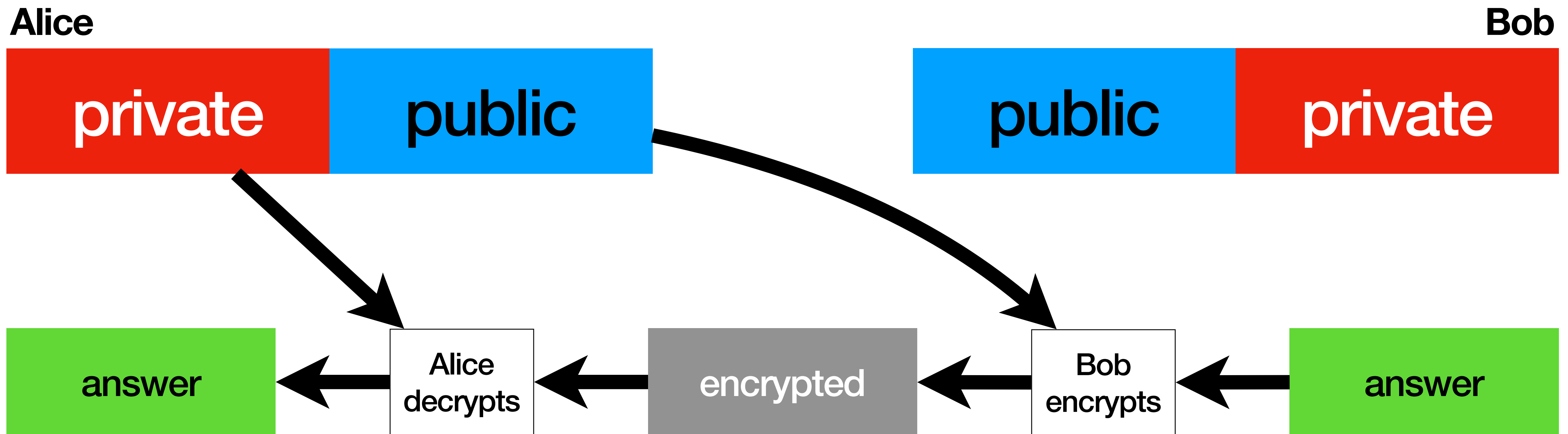
Bob sending a private answer to **Alice**

RSA private communication



Bob sending a private answer to **Alice**

RSA private communication

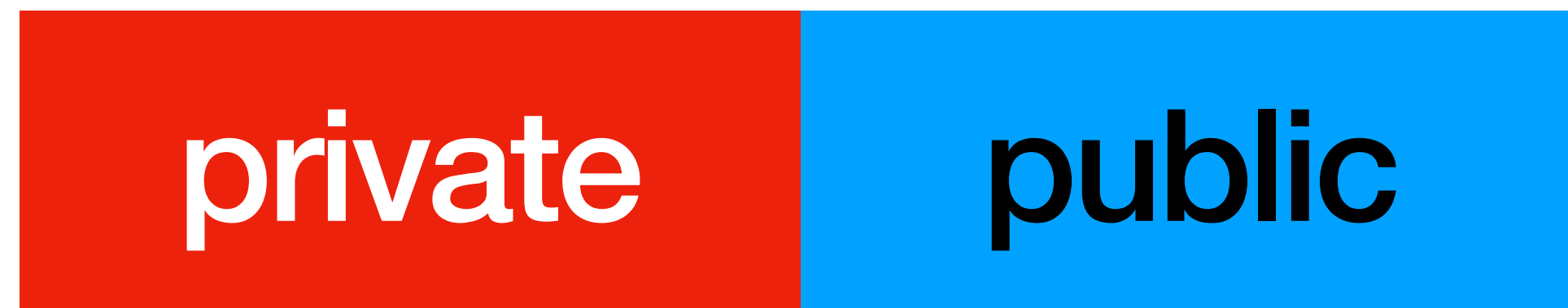


Bob sending a private answer to **Alice**

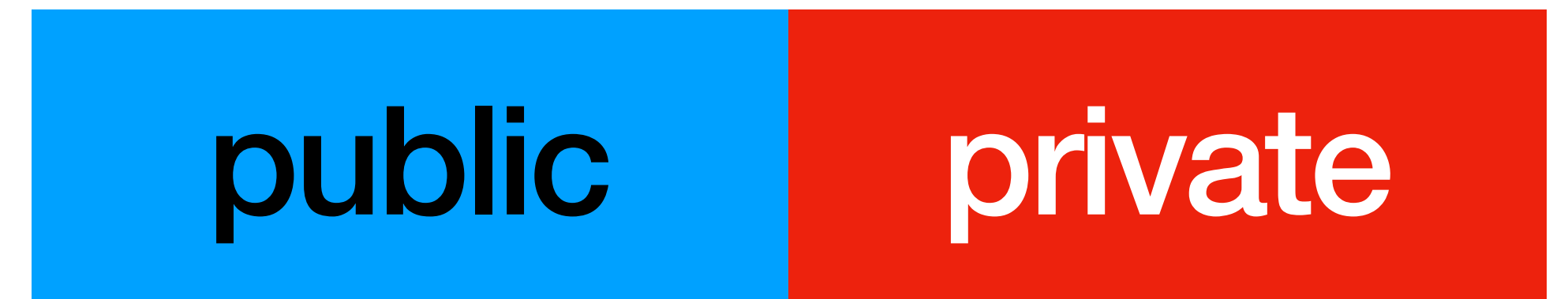
RSA signing / authentication

RSA signing / authentication

Alice

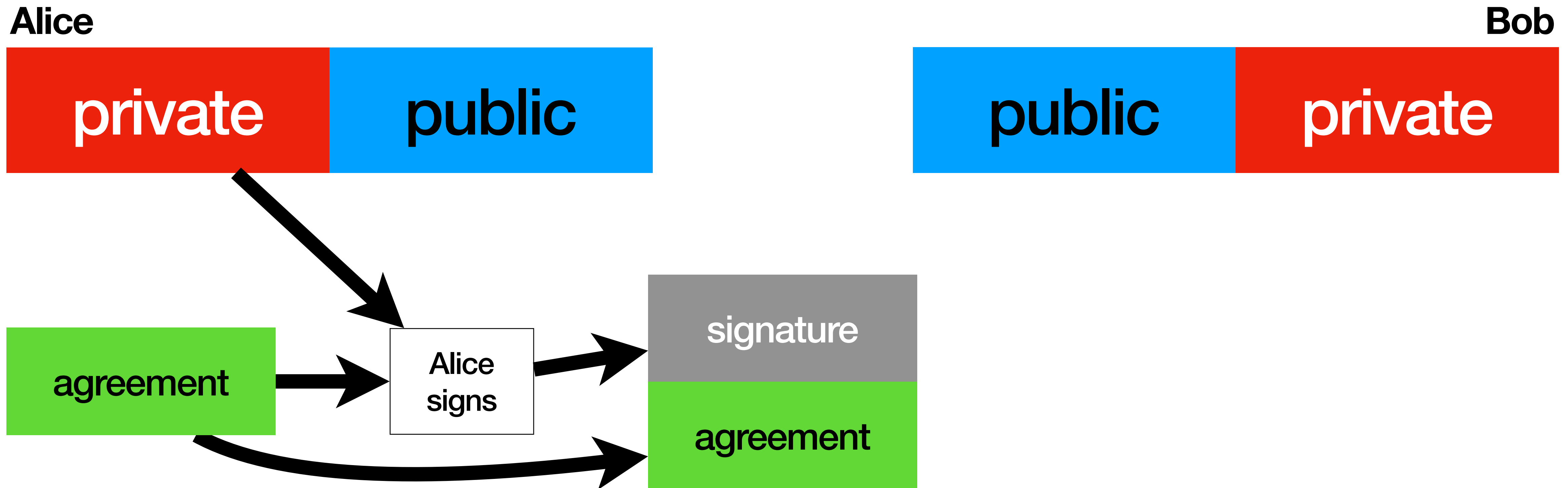


Bob



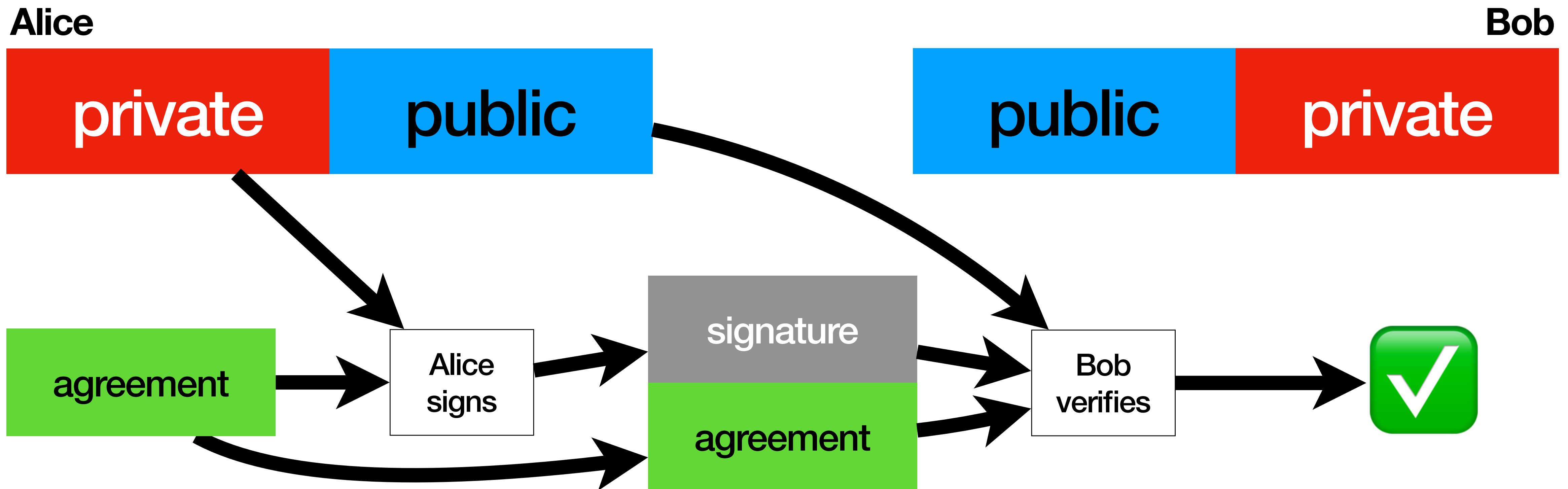
Alice sending an authenticated agreement to **Bob**

RSA signing / authentication



Alice sending an authenticated agreement to **Bob**

RSA signing / authentication

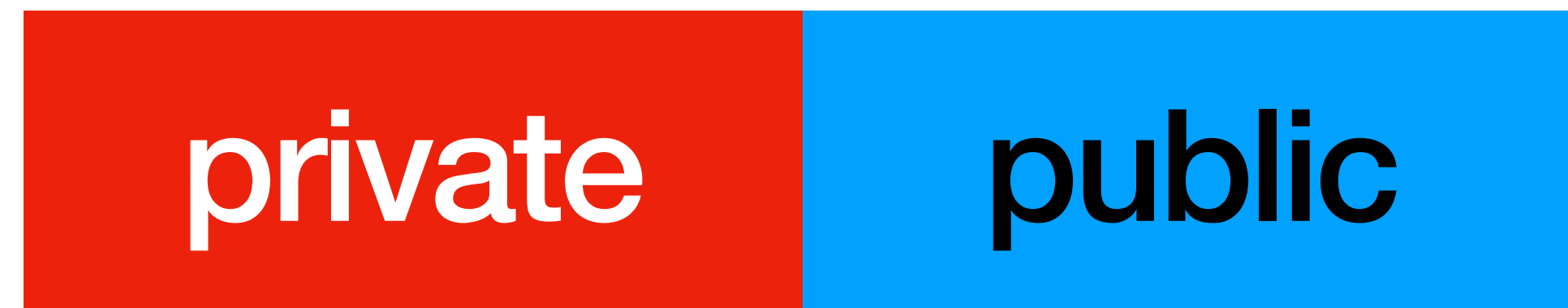


Alice sending an authenticated agreement to **Bob**

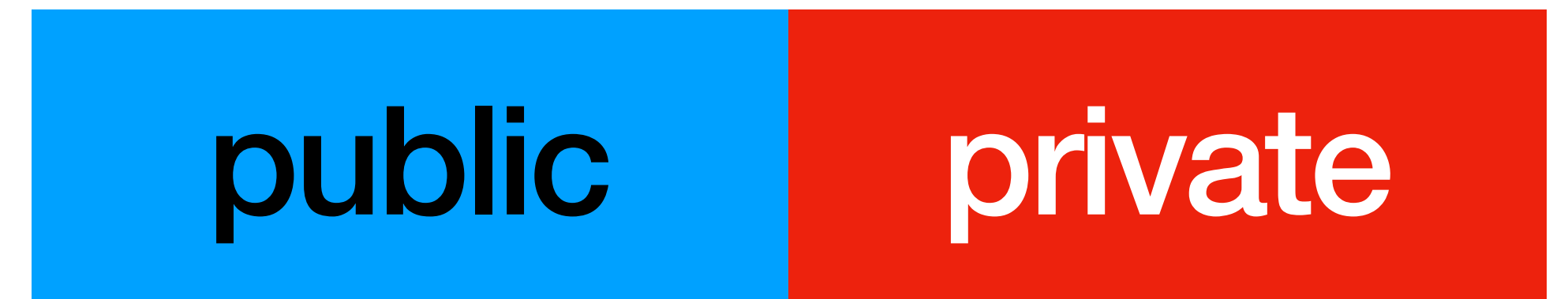
RSA private and authenticated

RSA private and authenticated

Alice

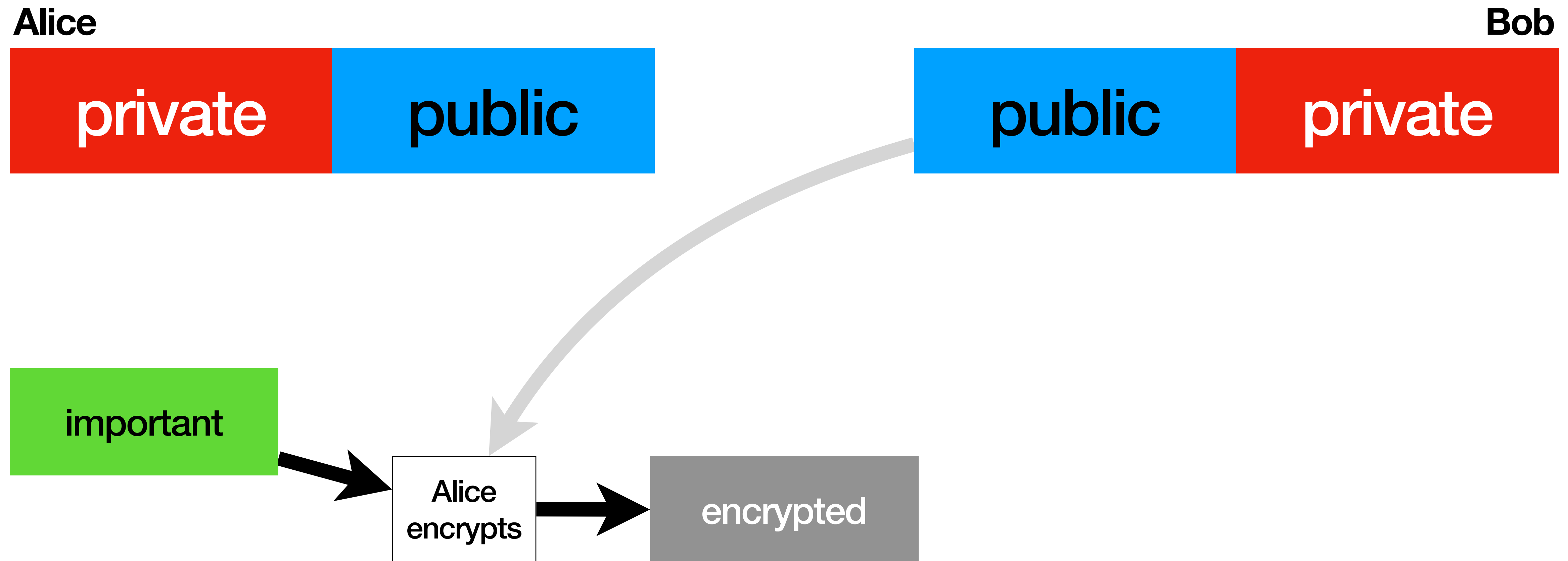


Bob

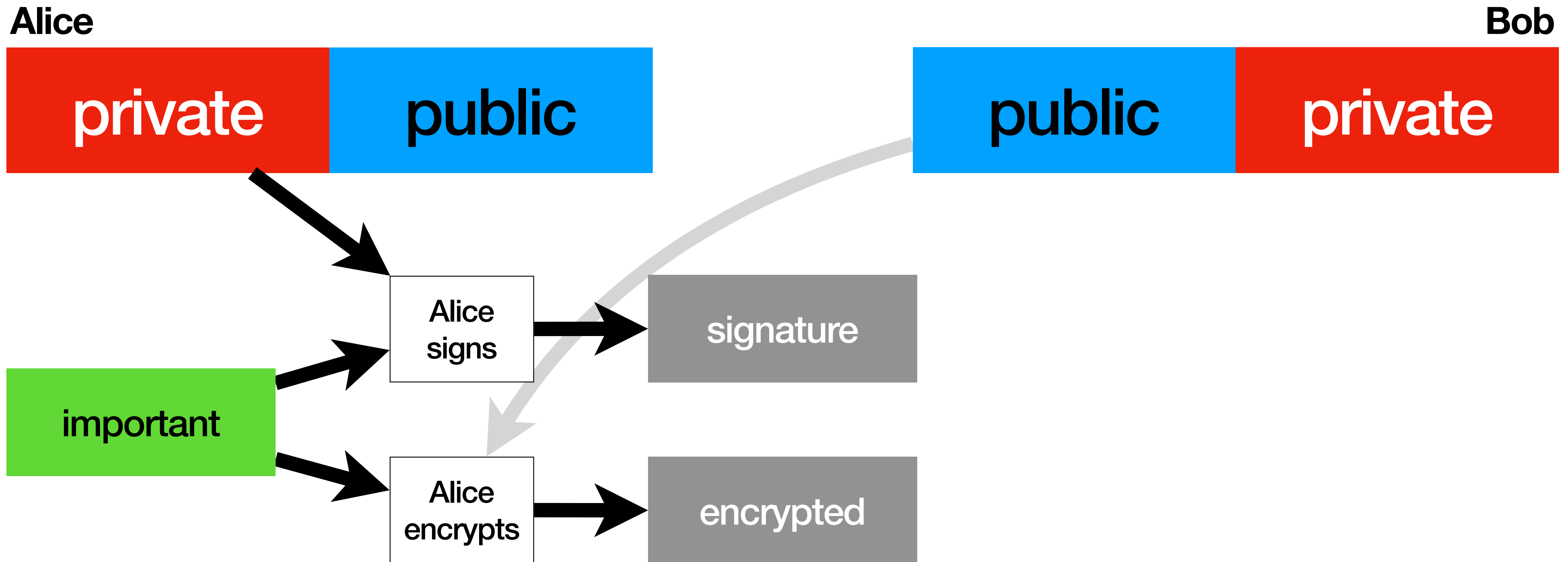


important

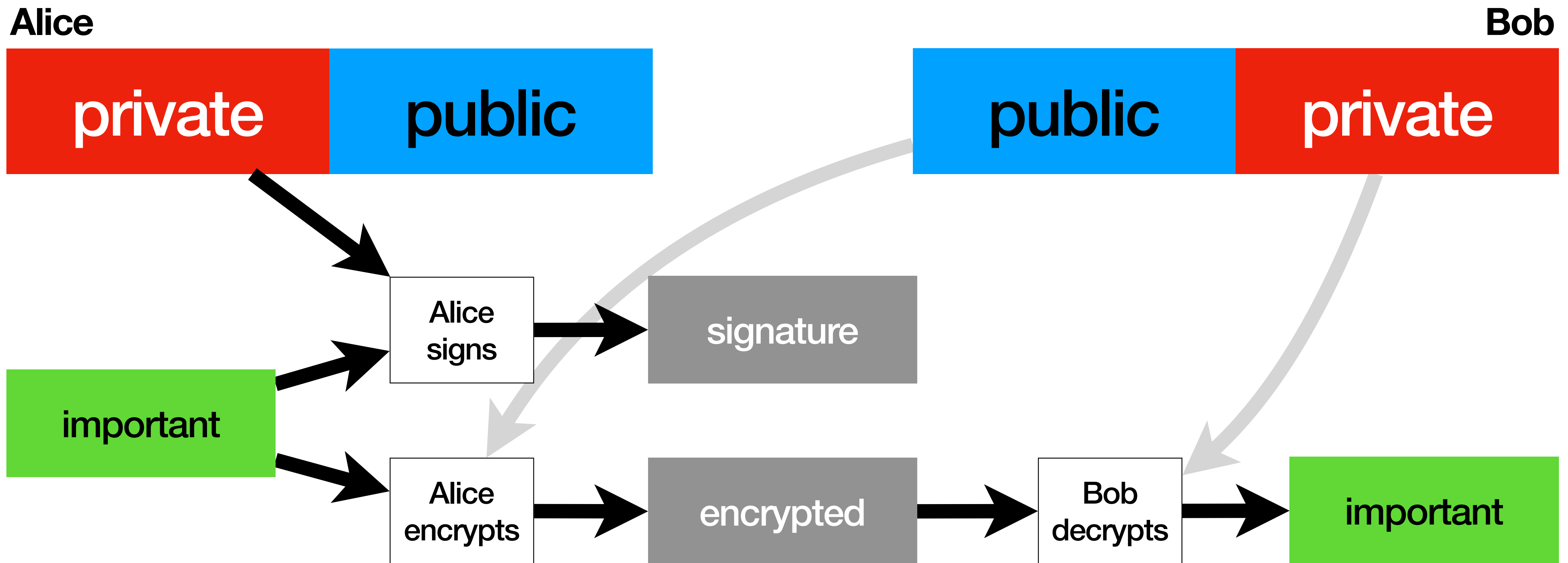
RSA private and authenticated



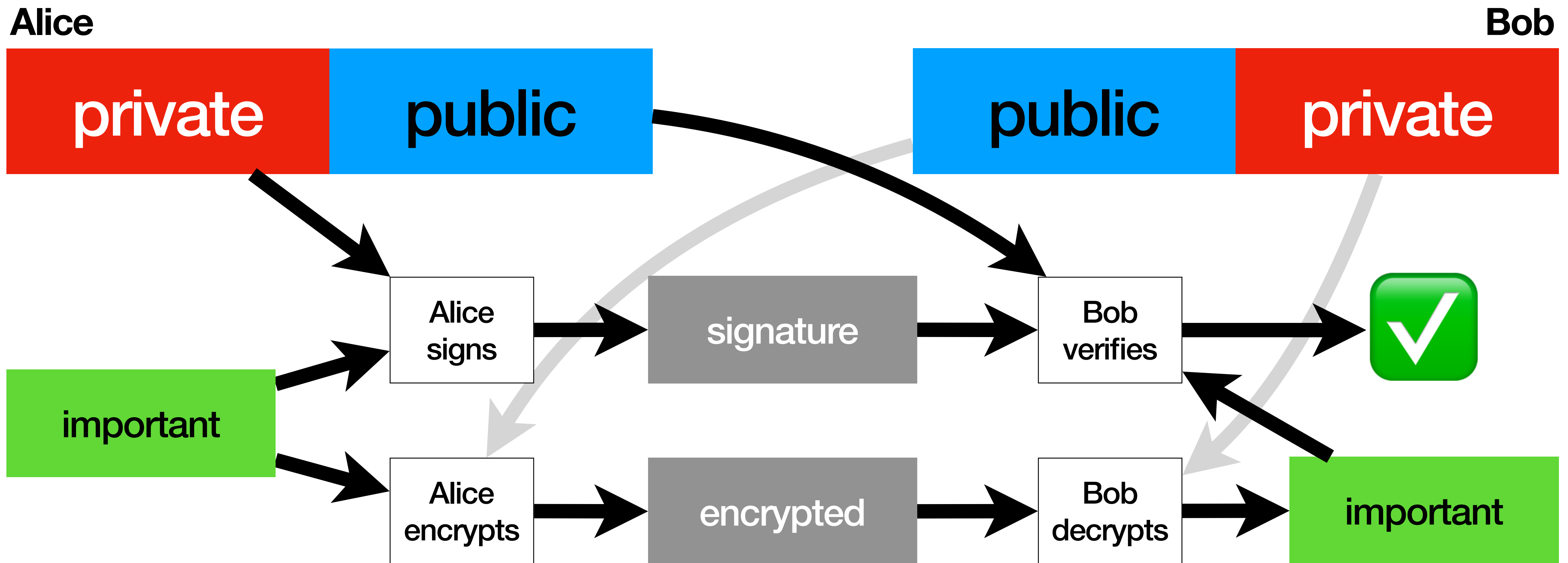
RSA private and authenticated



RSA private and authenticated



RSA private and authenticated



RSA

RSA

no way this is enough

X.509 Certificates

X.509 Certificates

issuer name

subject name

valid from

valid to

public key

signature

X.509 Certificates

issuer name
subject name
valid from
valid to
public key
signature

Issuer certifies,
by **signing** the document:

- This **public key**
- Belongs to this **Subject**
- During this **period**

X.509 Certificates

issuer name
subject name
valid from
valid to
public key
signature

the subject's

the issuer's

Issuer certifies,
by **signing** the document:

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X.509 Certificates

issuer name
subject name
valid from
valid to
public key
signature

the subject's

the issuer's

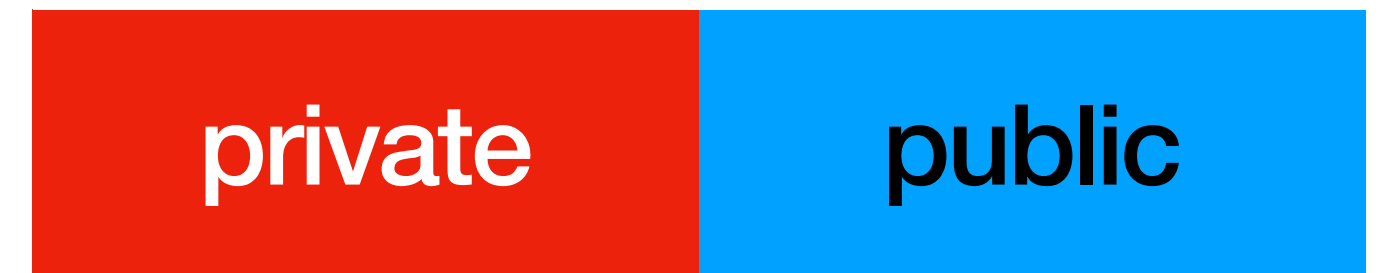
Issuer certifies,
by **signing** the document:

- This **public key**
- Belongs to this **Subject**
- During this **period**

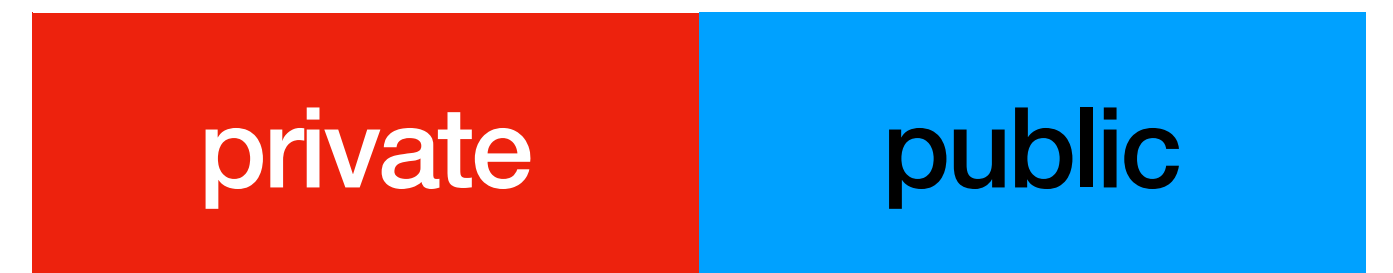
Certificates provide **Identity Assurance**

X.509 Certificates

Carol



Dave



Carol certifying this public key is **Dave's**

X.509 Certificates

issuer name	Carol
subject name	Dave
valid from	...
valid to	...
public key	
signature	

Carol

private

public

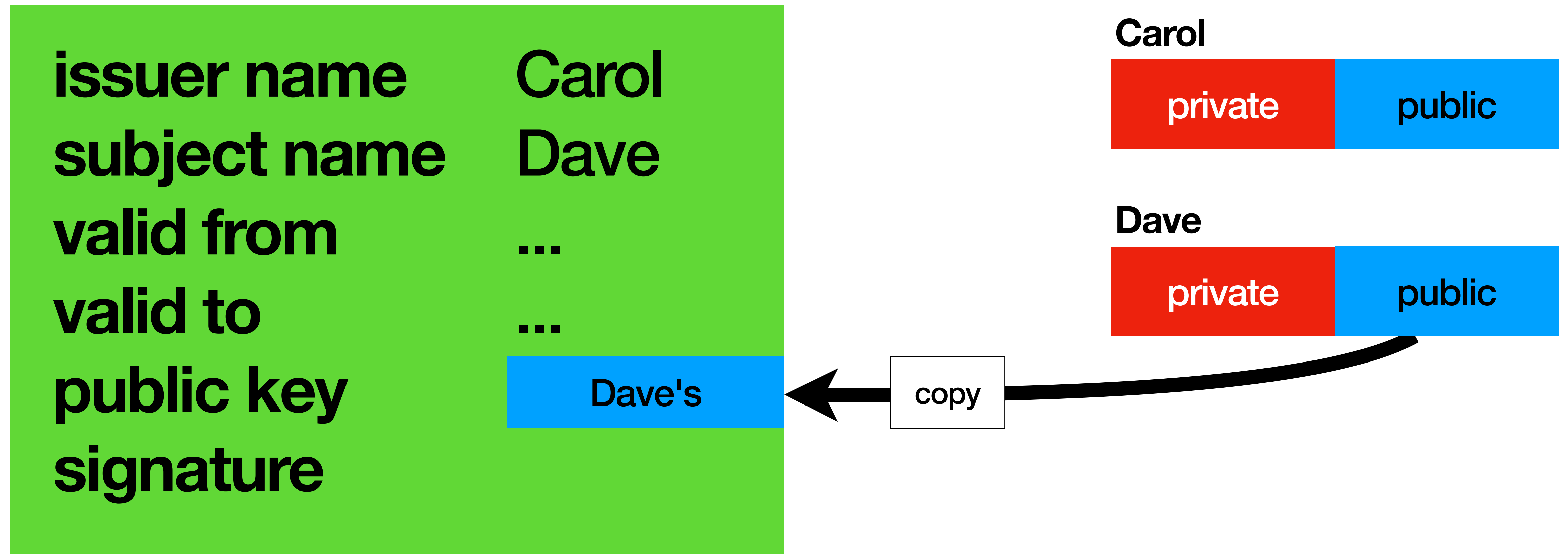
Dave

private

public

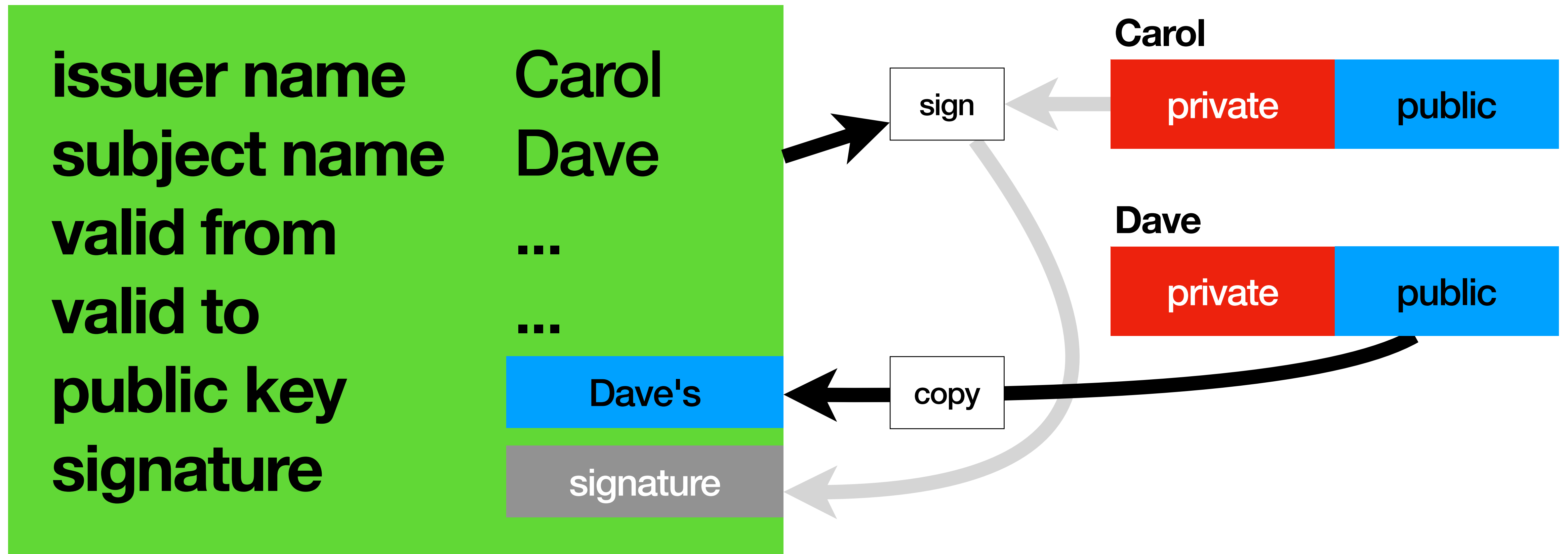
Carol certifying this public key is **Dave's**

X.509 Certificates



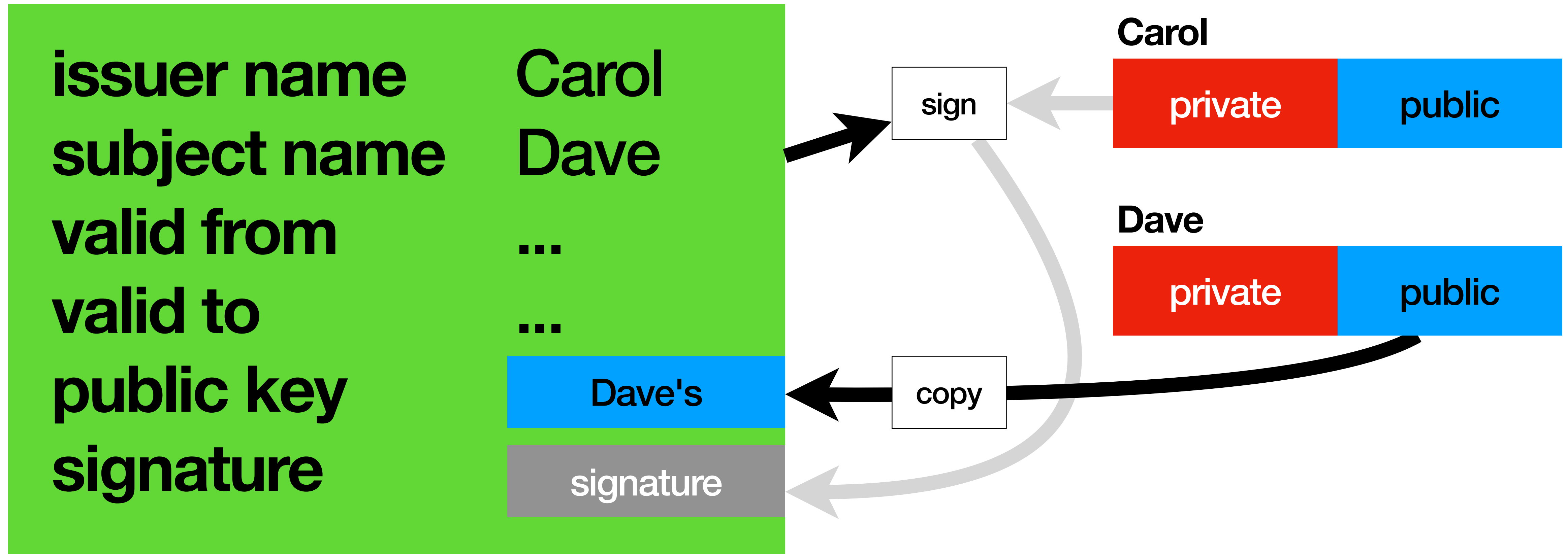
Carol certifying this public key is **Dave's**

X.509 Certificates



Carol certifying this public key is **Dave's**

X.509 Certificates



Certificates hold no secrets and are **public**.

X.509 Self-Signed Certificates

X.509 Self-Signed Certificates

issuer name	Bob
subject name	Bob
valid from	...
valid to	...
public key	
signature	

Bob

private

public

X.509 Self-Signed Certificates

issuer name	Bob
subject name	Bob
valid from	...
valid to	...
public key	Bob's
signature	

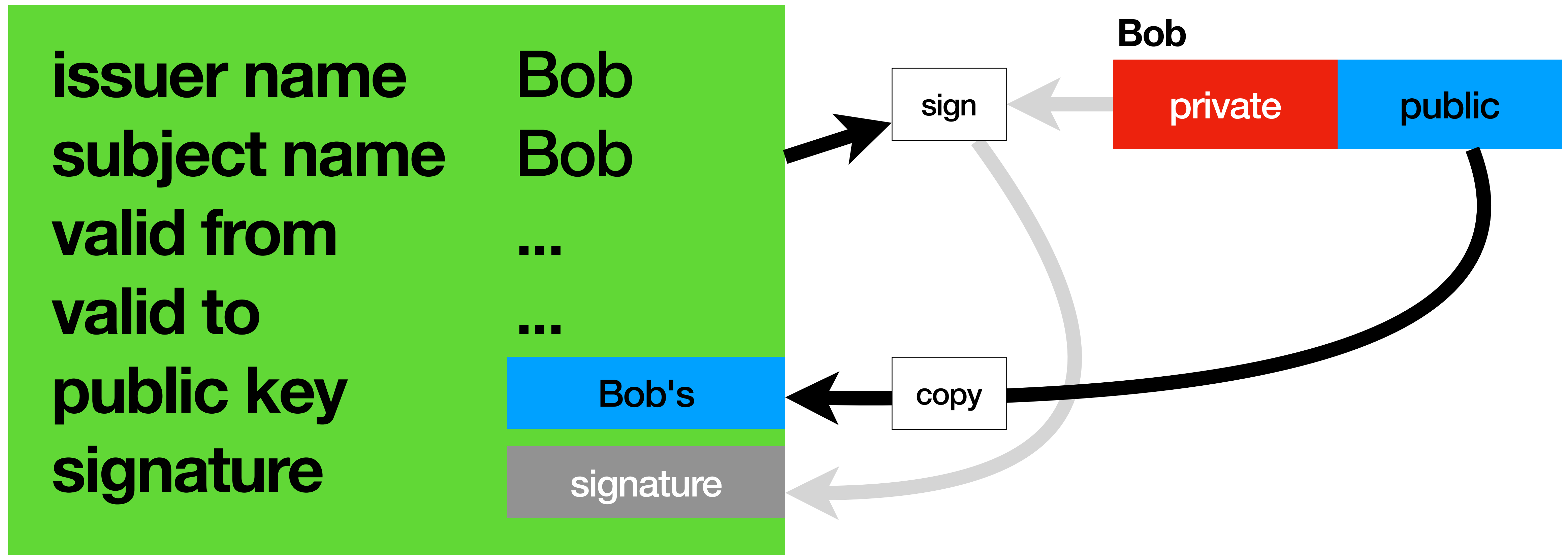
Bob

private

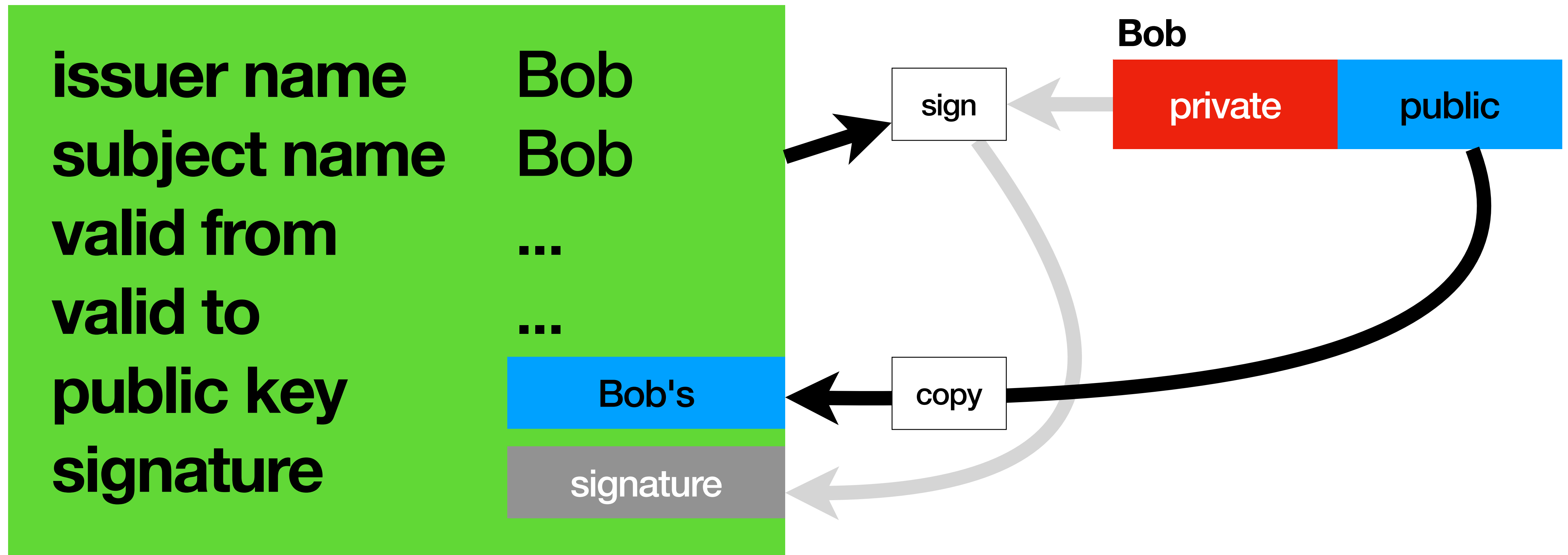
public

copy

X.509 Self-Signed Certificates



X.509 Self-Signed Certificates



Self-Signed Certificates are **auto proclamations of Identity**

Certificates

Certificates

no way self-signed certificates are enough

Certificate Authorities

Certificate Authorities

- Issue (actually **Sign**) **Certificates**

Certificate Authorities

- Issue (actually **Sign**) **Certificates**
- Using their own **private** key

CA

private

public

Certificate Authorities

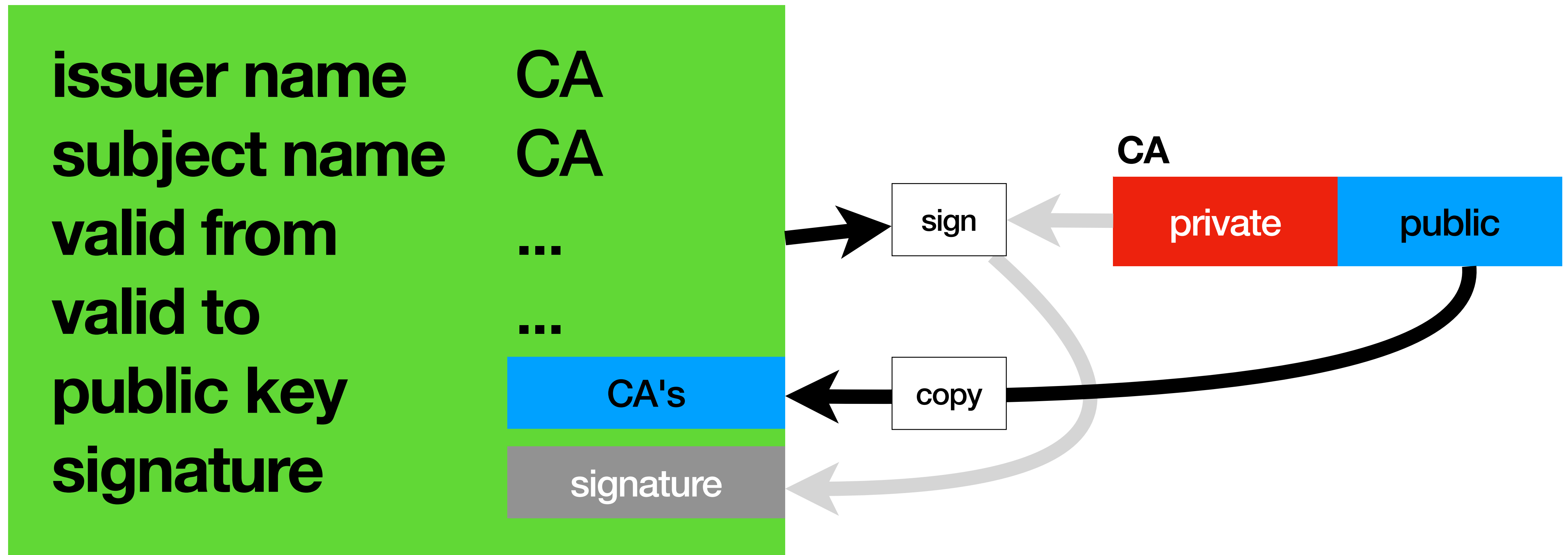
- Issue (actually **Sign**) **Certificates**
- Using their own **private** key
- And state their own **Identity** via a **Self-Signed Certificate**

CA

private

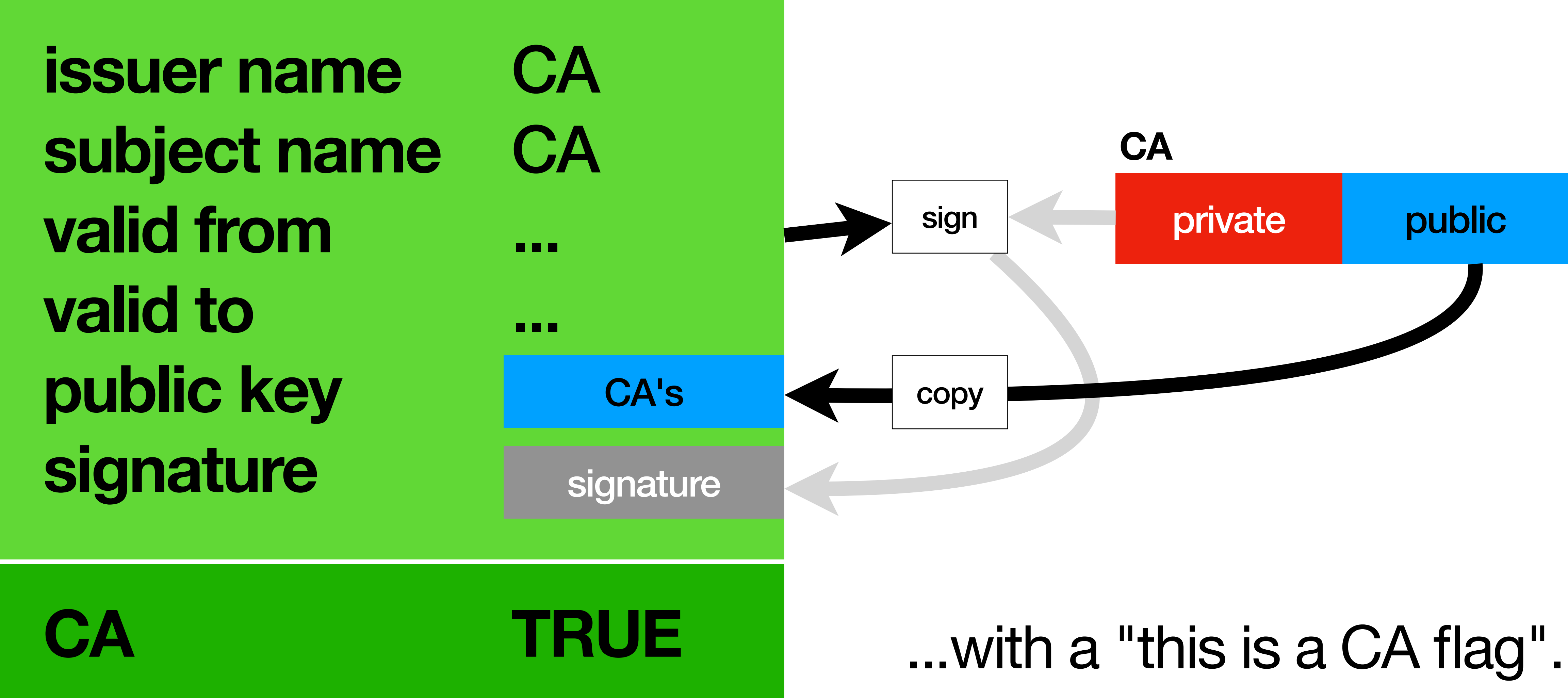
public

Certificate Authorities



Certificate Authorities are built on Self-Signed Certificates

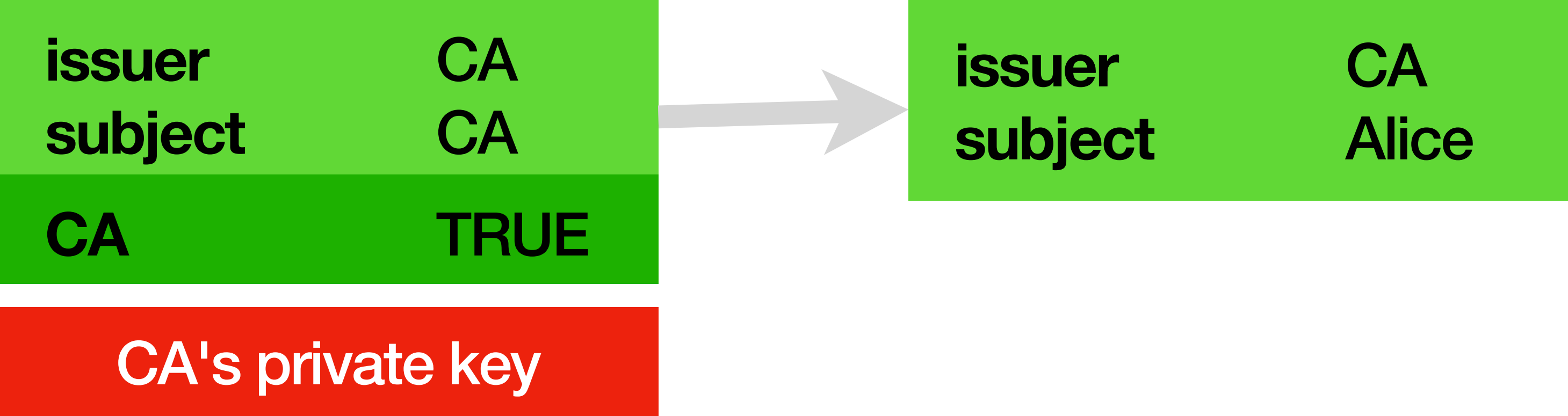
Certificate Authorities



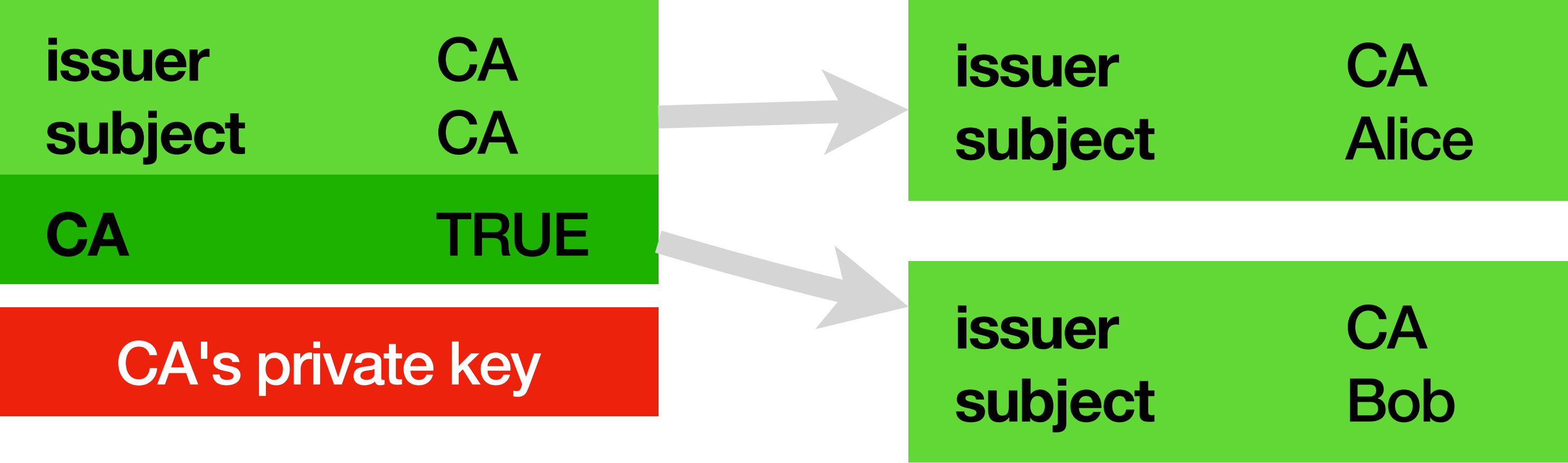
Certificate Authorities

issuer	CA
subject	CA
CA	TRUE
CA's private key	

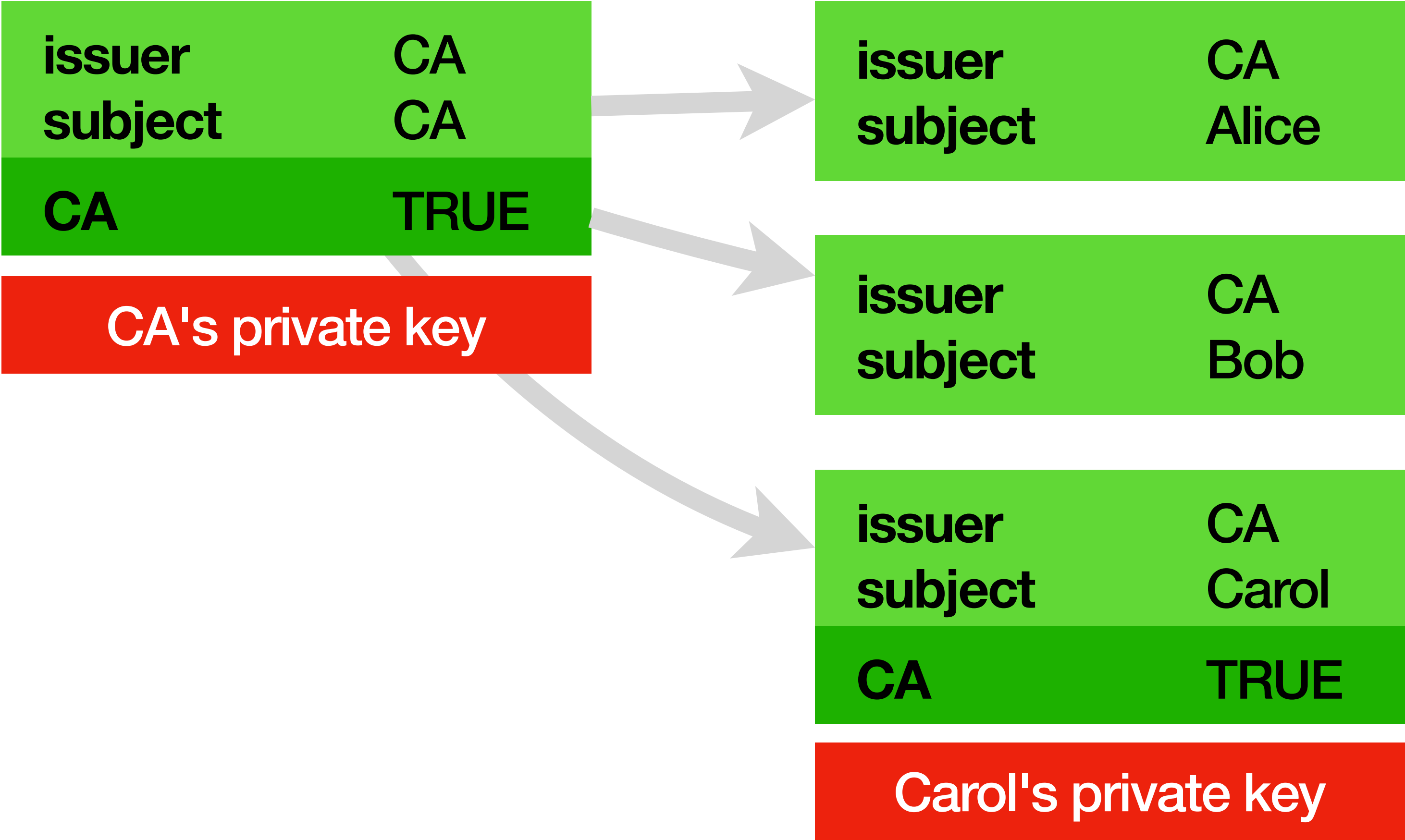
Certificate Authorities



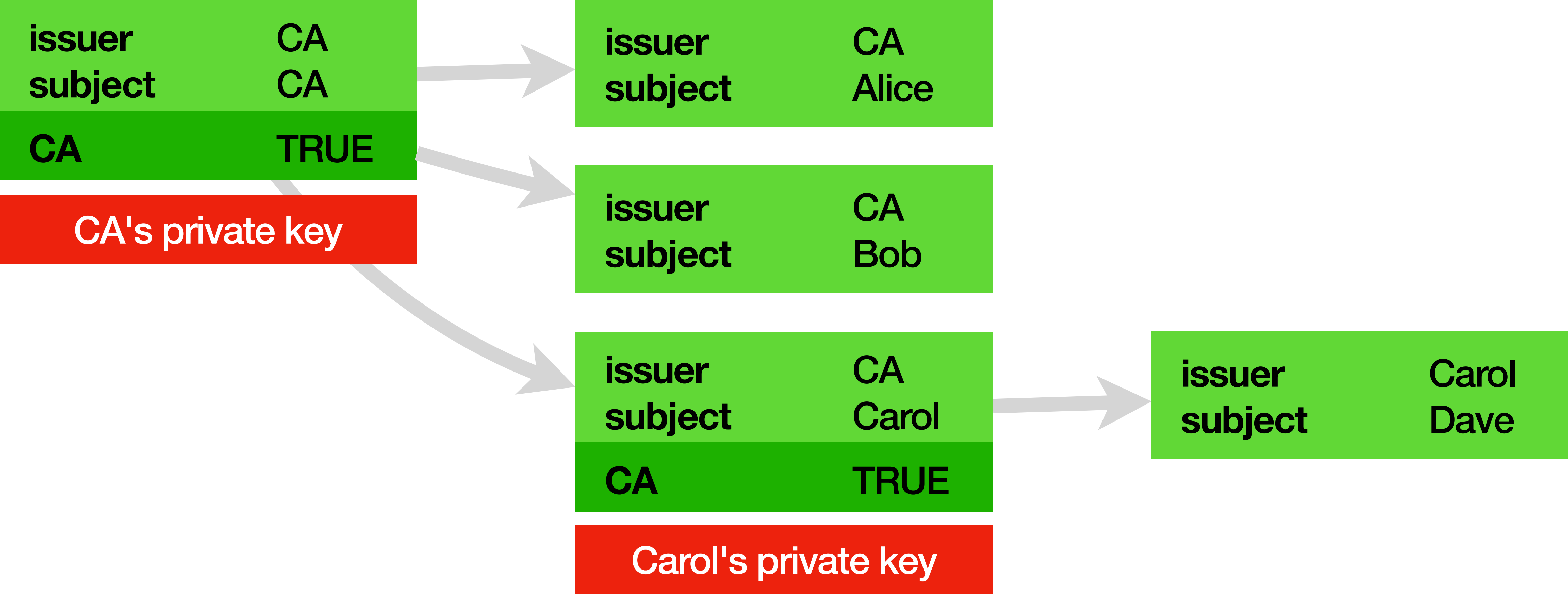
Certificate Authorities



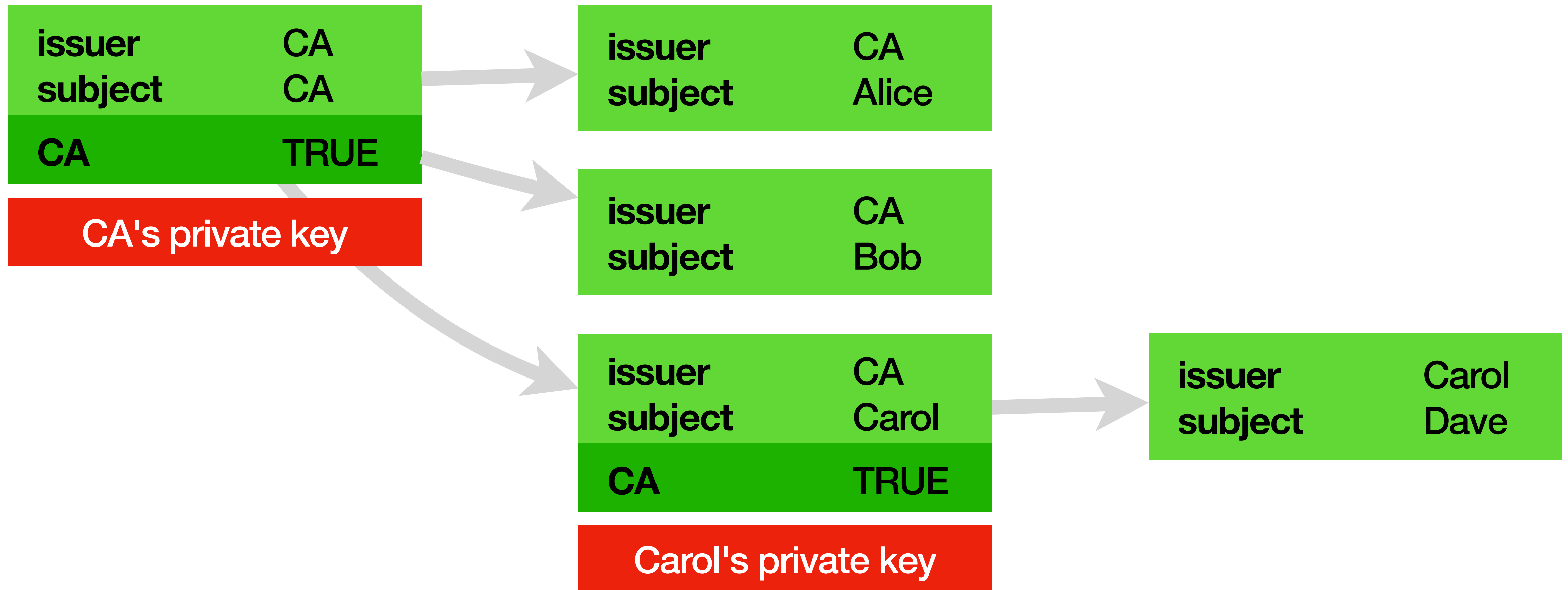
Certificate Authorities



Certificate Authorities



Certificate Authorities



...turtles all the way down.